Main files

Notebook

- `Notebook_A1_validated_sapphire_vortex_framework.ipynb`

This is the main Python/Colab notebook used to generate the simulations, diagnostic tables, and figures. It includes:

- material parameters,
- beam parameters,
- numerical grid settings,
- split-step Fourier propagation routines,
- nonlinear propagation operators,
- plasma/electron-density evolution,
- spectral and fluence diagnostics,
- plotting routines.

Output tables

The `outputs/` directory contains processed simulation results used in the manuscript:

File	Description
`fused_silica_vortex_validation.csv`	Fused-silica vortex validation scan
`sapphire_ordinary_vortex_scan.csv`	Ordinary-sapphire vortex scan
`sapphire_extraordinary_vortex_scan.csv`	Extraordinary-sapphire vortex scan
`combined_material_comparison.csv`	Combined comparison table for fused silica and sapphire

Typical columns include:

- `material`
- `beam_type`
- `energy_nJ`
- `peak_power_MW_est`
- `Imax_TW_cm2`

- `z_Imax_mm`
- `final_cutoff_nm`
- `energy_loss_pct`
- `max_rho_cm3`
- `runtime_s`

Figures

The `figures/` directory contains the final manuscript figures:

File	Description
`Fig1.png`	Fused-silica vortex validation
`Fig2.png`	Fused-silica S-scan / spectral broadening proxy
`Fig3.png`	Representative sapphire ordinary vortex propagation sequence
`Fig4.png`	Material comparison: fused silica vs sapphire
`Fig5.png`	Sapphire ordinary vs extraordinary polarization comparison

Model overview

The simulation uses an x, y, t, z propagation framework for femtosecond pulses in transparent nonlinear media. The model includes:

- diffraction,
- group-velocity dispersion,
- Kerr self-focusing,
- multiphoton absorption,
- plasma defocusing,
- electron-density evolution.

The initial field is a focused $m = 1$ vortex pulse of the form

$$E(r, \phi, t, z = 0) = E_0 \left(\frac{r}{w_0}\right)^{|m|} \exp\left(-\frac{r^2}{2w_0^2}\right) \exp(im\phi) \exp\left(-\frac{t^2}{2\sigma_t^2}\right) \exp\left(-i\frac{kr^2}{2f}\right),$$

where $m = 1$, w_0 is the vortex beam radius, and f is the focusing-lens focal length.

Main simulation parameters

The main simulations use:

Parameter	**Value**
Central wavelength	800 nm
Pulse duration	65 fs
Vortex charge	$m = 1$
Vortex beam radius	0.060 mm
Focal length	15 mm
Grid size	$N_x, N_y, N_t = 192, 192, 256$
Transverse spacing	$dx = dy = 3.125\ \mu\text{m}$
Temporal spacing	$dt \approx 0.703$ fs
Temporal window	180 fs
Propagation distance	18 mm
Propagation step	$100\ \mu\text{m}$

Materials

The simulations include:

Material	n_0	n_2 (m^2/W)	U_i (eV)	K	k_2 (s^2/m)
Fused silica	1.4533	3.2×10^{-20}	9.0	6	36×10^{-27}
Sapphire ordinary	1.7600	3.0×10^{-20}	8.8	6	70×10^{-27}
Sapphire extraordinary	1.7520	3.0×10^{-20}	8.8	6	70×10^{-27}

Some sapphire ionization and plasma parameters are approximate or proxy values. Therefore, the sapphire thresholds reported in the manuscript should be interpreted semi-quantitatively.

Reproducing the results

The notebook can be run in Google Colab or locally in a Jupyter environment.

Option 1: Run in Google Colab

1. Upload or open `Notebook_A1_validated_sapphire_vortex_framework.ipynb`.
2. Run the setup/import cells.
3. Run the material and grid definition cells.
4. Run the fused-silica validation and sapphire scan cells.
5. Generate tables and figures.

Full propagation scans may require substantial runtime because each energy case performs a full x, y, t, z simulation.

Option 2: Inspect processed outputs

The processed CSV files in `outputs/` contain the data used to generate the manuscript tables and comparison plots. These files allow verification of the reported trends without rerunning the full simulations.

Numerical bandwidth note

The temporal grid imposes a finite spectral bandwidth. For the main grid, the estimated minimum resolved wavelength is approximately $276\ \mathrm{nm}$. Blue-side cutoff wavelengths approaching this value should be interpreted as bandwidth-limited indicators of strong spectral broadening rather than fully resolved physical cutoff wavelengths.

This limitation mainly affects the most strongly broadened fused-silica cases.